# Week 9 web page: delivery guide

Instructor notes for Week9_MCMC.html

## General

- Page: https://kevintshoemaker.github.io/NRES-746/Week9_MCMC.html. Open it before class; the widgets need no internet once loaded.
- For live demos, run the matching section of `CODE_Week9_MCMCAndStan.R` (same seeds, same numbers). Run the first part before the Wednesday sections: they reuse `nll_gamma()`, `plot_surface()`, `log_post_germ()`, and `metropolis()`. Compile the Stan model before class (up to a minute the first time).
- Zoom the browser to 125-150% for the projector.
- Questions on the page have no answers. Answers are below; *Land:* is where discussion should end up.
- Readings for Canvas: Bolker 7.2.3-7.3 (printed pp. 302-319); optional: Monnahan et al. 2017 (MEE 8:339-348), *Statistical Rethinking* Ch. 9.
- No in-class WODs this week. Take-home WOD *Getting Stan Running* goes out at the end of the Wednesday period (due Monday); hand out the cmdstanr quick start (`NOTES_Reference_CmdstanrQuickStart.pdf`) with it.

## As taught (Oct. 19-21, 2026)

- **Monday** covered sections 1-4 of the first part (Nelder-Mead, annealing, Metropolis) with no WOD. The closing discussion (three germination chains) wasn't reached; it opened Wednesday as a debrief after Lab 7 Act 1 Part 2 had students judge chains first. It now sits in the page's third part as "Which chains can you trust?" Per the post-lecture announcement (added Nov. 4): the page now says Nelder-Mead is `optim()`'s deterministic default (`#nelder-mead`), that annealing can pass over local minima but isn't guaranteed to (`#annealing`), and that SANN and MCMC differ only in aiming for a point estimate vs. the joint posterior (top of `#metropolis-rule`). Chi Feng's MCMC gallery (shared by a student, via Perry Williams's Bayesian class) is now also linked at the end of `#metropolis-rule`. Reading for Tuesday: the MCMC section of Bolker Ch. 7.
- **Lab 7:** most groups got 45+ min on Act 2, which went home.
- **Wednesday:** the debrief, then HMC and Stan; it went well. While showing Chi Feng's sampler gallery, the instructor said Gibbs is "always best with few parameters." That was corrected afterward (the issue is correlation, not the number of parameters); the page's Gibbs section states the corrected version. Per the post-lecture announcement

(added Nov. 4): the page now also explains why Gibbs looked good on the gallery's banana (symmetric, axis-aligned, overall correlation near zero; the gallery has no strongly diagonal target), adds the energy-conservation framing, the gradient as a vector of partials, and the typical set to `#hmc-algorithm`, and opens `#stan` with the sampler's ingredients (data, parameter space, likelihood, priors, optional inits) and four reasons to use Stan (efficient sampling, compiled C++, scales to thousands of parameters, informative warnings). `#hmc-failures` adds that discrete latent variables just need reparameterizing (the instructor hopes to fit one in Stan later). Assigned: the Stan take-home WOD (install and run Stan), Bolker Ch. 6 model comparison section, optional Monnahan et al. 2017 (on WebCampus, not linked from the page).

- The page was built after the week from the PDF NOTES (now in `lecture_notes/archive/`) and the legacy LECTURE7 page; Gibbs, thinning, the funnel, and the widgets are new.

# Part 1: Derivative-free optimization and Metropolis (Monday, ~35 min)

### Big picture (~4 min)

- Point at FIT. Derivative-based methods (Newton, BFGS) are fast near the answer but need slopes. Two questions on the page; the second grows out of the first.

### Nelder-Mead (~7 min)

- Draw one triangle and its moves on the board (the page figure).
- Run the triangle plot; then the widget, default start ("far up the valley wall"): 10 steps, best $f$ 2.60 (22 evaluations); 30 steps, 0.015; 60 steps, $7 \times 10^{-10}$ (113 evaluations). Point at the long expansion down the wall, then the crawl along the curve.
- `optim()` default: 147 evaluations, NLL 37.67.
- "Why do the triangles keep shrinking, and how could you use their size to decide when to stop?" *Land:* near the minimum every reflection overshoots, so contractions and shrinks dominate; the triangle's size tracks how well the minimum is located. `optim()` stops when the NLL values at the vertices nearly agree (`reltol`).

### Simulated annealing (~8 min)

- Read the acceptance table: at $T = 1$ an NLL 2 units worse is accepted 14% of the time; at $T = 0.1$, essentially never.
- Widget, in this order (predict before each run):
  - Default ($T = 2$, cool $\times 0.85$): finds the deep valley (best $x = -2.10$, 100% of the last 1,000 steps there).
  - Cold start (slider far left, $T \approx 0.05$): stuck in the shallow valley (0%, best $x = 1.89$).
  - Hot and never cooling (cooling = 1): wanders between both (82% in the deep one); the histogram matches the orange $e^{-f/T}$ curve. Say: "hold that thought" (it's Metropolis at $T = 2$).

- Myxo annealer: 3,000 evaluations vs. Nelder-Mead's 147.
- "SANN reports 40.77 with `convergence = 0`; Nelder-Mead reached 37.67. Broken?" *Land:* untuned, not broken. SANN runs a fixed 10,000 iterations and always reports 0. Use a global search to find the right valley, then polish with Nelder-Mead or BFGS.

**Metropolis (~15 min)**

- Board: the three changes, then Bayes' rule in the ratio; $P(y)$ cancels. Then the $A/B$ argument (1 min).
- Stress: a rejection records the current value again; the chain should wander, not converge.
- Germination run: 59% accepted; mean 0.734 vs. exact 0.738; histogram matches Beta(31, 11).
- "The annealer used `runif(1) < exp(-delta / temp)`; `metropolis()` uses `log(runif(1)) < lp_prop - lp_current`. Different rules?" *Land:* no. $u < e^x \iff \log u < x$; with $T = 1$ and the NLL difference replaced by the log posterior difference, both lines make the same decision.
- Myxo chain: 28% accepted; shape interval 29 to 81. Mention it mixes slowly; don't explain why yet.
- **Behind?** Skip the SANN question and the myxo chain.

**Before the lab: don't show** the page's third part (chain checks, random walks). Lab 7 Act 1 Part 2 asks students to judge chains, and Act 2 2c asks why centering fixes a badly mixing sampler.

# Part 2: Working on the log scale (Lab 7 opening, ~5 min)

- Run the underflow chunk: `prod()` gives 0, the sum of logs gives $-2147$.
- Log posterior = log likelihood + log prior; accept if $\log u <$ the difference; record the current value either way.
- Examples only with the germination data, never the lab's numbers. Act-by-Act answers: `labs/answer_keys/Lab7_Act1_Key.pdf`, `Lab7_Act2_Key.pdf`. Common Act 2 bugs: not updating the current log posterior after an acceptance; storing only accepted steps; the same seed for both runs.

# Part 3: Chains, HMC, and Stan (Wednesday, ~40 min)

**Which chains can you trust? (~6 min)**

- Show the three-chain table first (accepted 87%, 59%, 4%; means 0.38, 0.73, 0.70), then ask the page question before the trace plots.
- "A labmate says the first chain is the best, since nearly every proposal was accepted." *Land:* the middle one. Tiny steps are accepted but barely move (still in burn-in); huge

steps are rejected, so the chain sits still and its mean is close by luck. Check the trace plot, drop burn-in, run several chains, and tune the step size. Acceptance is not a quality score.

- Name the terms: burn-in (Stan: warmup), mixing, R-hat, ESS. Map to Lab 7 Act 1: tiny steps = Chains B and D; huge = Chain C.
- Widget walkthrough (seed fixed until "New random numbers"):
  - Default (SD 0.1, one chain): 59%, mean 0.731, ESS 62. Burn-in 100: mean 0.741, ESS 220.
  - SD 0.005: 88% accepted, mean 0.404, ESS 3. Turn on three chains: R-hat 2.38.
  - SD 3.16 (far right): 2% accepted, ESS 6; three chains, R-hat 1.22.
  - Back to SD 0.1, three chains, burn-in 100: R-hat 1.00, ESS 702.
- ACF figure: germination ESS 914 of 4,500 kept steps; myxo shape 43 of 18,000. Thinning: fewer effective draws, not more.

**Why random walks struggle (~7 min)**

- Board: a tilted ellipse with small circle proposals; distance grows like $\sqrt{\text{steps}}$.
- Random walk on $\rho = 0.95$: ESS 6 of 200. Gibbs: ESS 14.
- Gibbs: exact conditional draws, no rejections, but moves only along the axes by the conditional SD $\sqrt{1-\rho^2}$ (0.31 at 0.95). The table: ESS per sweep 0.60, 0.05, 0.01 at $\rho$ = 0.5, 0.95, 0.99. Correlation, not the number of parameters, is what matters.

**HMC (~10 min)**

- Board: the puck (kick, slide, accept on total energy). It's still Metropolis, with far-reaching proposals that are nearly always accepted.
- Energy: $H = -\log P(\theta \mid y) + \frac{1}{2}\sum m^2$; accept with $\min(1, e^{-\Delta H})$. Exact physics conserves $H$; the accept/reject step corrects the leapfrog error. Same form as the annealing rule at $T = 1$, applied to total energy.
- ESS: random walk 6, Gibbs 14, HMC 183 of 200.
- "Each HMC iteration costs 11 gradient evaluations. Still worth it?" *Land:* yes: effective draws per evaluation are 6/200 = 0.03 for the random walk vs. 183/2,200 = 0.08 for HMC, and the gap grows quickly with more parameters.
- Sampler widget ($\rho = 0.95$, defaults; seed fixed):
  - Random walk: 67% accepted, lag-1 0.90, ESS 16.
  - Gibbs: lag-1 0.89, ESS 7. Slide $\rho$ to 0.5: ESS 134; to 0: 198; to 0.99: 3.
  - HMC ($\epsilon$ 0.1, $L$ 10): 99% accepted, ESS 51, 2,200 evaluations. $L = 2$: ESS 10 (random-walk-like). $L = 30$: ESS above 200 (trajectories cross to the far side each time). This sensitivity is why NUTS picks $L$.
- If asked: NUTS doubles the trajectory until a U-turn, so drawn trajectories look like arcs; $\epsilon$ is the leapfrog time step, tuned in warmup to `adapt_delta` (0.8). The default mass matrix is diagonal (fixes scale, not correlation). Autodiff applies the chain rule mechanically: exact gradients, no hand derivation.

**Stan (~10 min)**

- Blocks table; `~` adds to `target` (the log posterior, not its negative; HMC's potential energy is $-$`target`).
- Output: R-hat 1.00, ESS 452 for shape (vs. 43), divergences 0, shape 95% interval 25 to 76 (Metropolis: 29 to 81).

**When HMC struggles (~4 min)**

- Funnel figure: centered vs. non-centered. Divergences are a bias warning, not just slowness. Gibbs/JAGS doesn't diverge in a funnel (no global step size), but non-centering helps Stan far more than JAGS (Monnahan et al.: 2.8 to 129 times faster in Stan).
- **Behind?** Skip the mass-matrix and autodiff bullets and the funnel.

**Discussion (~5 min)**

- Q1 (Stan overkill?) *Land:* compare effective draws per second, not steps per second: about 43/0.12 s for Metropolis vs. 452/0.3 s for Stan; compiling happens once per model. For two parameters either is fine; Stan earns its keep as models grow and when diagnostics matter.
- Q2 (when Stan, when JAGS?) *Land:* Stan for many correlated parameters and hierarchical models; JAGS or NIMBLE for discrete latent states (occupancy, mark-recapture) unless you marginalize.
- Q3 (Gibbs accepts everything) *Land:* acceptance says nothing about how far a move goes; each Gibbs move is axis-parallel and as wide as the narrow conditional, so the chain zigzags (lag-1 autocorrelation $\rho^2$).